
An Ensemble Learning Framework with Explainable AI for Intelligent Pet Adoption Outcome Prediction in Animal Shelters

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Abstract:

Animal shelters face consistent problems such as overcrowding, lack of resources, and poor adoption matching. These issues contribute to higher euthanasia rates due to traditional intake procedures where human assessments play a critical role without any consistency or the use of multivariate data collected using contemporary shelter management systems. In our research, we develop an EEML approach combining gradient-boosted algorithms like XGBoost, LightGBM, and CatBoost with a stacking meta-learner to forecast the adoption result for sheltered dogs and cats (adopted, transferred, returned to owners, or euthanized). Our analysis uses a dataset with 148,317 clean shelter intake records from three municipal shelters located in different continents. There are 34 predictor variables related to an animal's age, sex, physical conditions, behaviour, as well as the shelter itself. Training and hyperparameters were selected using stratified five-fold cross-validation and Bayesian search procedure, respectively. Our ensemble method yielded good results, with a macro-averaged F1-score of 0.847. Also, the AUC-ROC metric showed the highest performance at 0.931, outperforming single models by 6.3–18.4 percentage points. Interpretability is provided through SHAP method; length of stay, type of intake, and age-at-intake became the most important features. The explainability layer enables the decision to be audited and acted upon by shelter personnel who lack machine learning knowledge. This makes the EEML approach a useful decision-making tool that is able to add real value in improving animal welfare within shelters.

Key Words: Animal Shelter Informatics; Ensemble Learning; XGBoost; LightGBM; SHAP Explainability; Adoption Outcome Prediction; Multiclass Classification

1. Introduction

In aggregate, international shelter systems process millions of dogs and cats per year. Nevertheless, organizational shortcomings with respect to intake classification, resource allocation, and adoption matching still account for

excessively high euthanasia rates. The American Society for the Prevention of Cruelty to Animals (ASPCA) estimates that there are approximately 6.3 million companion animal intakes in U.S. shelters per year, out of which some 920,000 animals are euthanized, not solely for medical reasons but largely as a consequence of overcrowding and unsuccessful placements [1]. Similar statistics can be reported regarding animal shelters in Europe and Asia-Pacific [2]. A recent technological advance in this sector involves the introduction of digital shelter management software, which makes it possible to track animals' activity in real time during the entire process from intake to placement. However, little effort has been made by the academic community towards exploiting shelter data to predict animals' future outcomes. The earliest computational methods of modeling adoption included straightforward logistic regressions or rule-based classifiers that could not account for the non-linear effects of predictors on outcome probability [3]. Further advancements in this field entailed using random forests and support vector machines, which provided higher classification accuracies; however, they generally worked well only on small-scale single-shelter datasets [4, 5]. Three main deficiencies characterize research conducted to date. One issue is the lack of interpretable results, because shelter managers are interested in understanding why certain actions are recommended based on predictions; however, this objective conflicts directly with the opaque nature of many efficient machine learning algorithms [6]. Another shortcoming lies in the heterogeneity of intake procedures, breed classification, and outcome terminology in multiple shelters, which hampers generalization [7]. Finally, the multiclass nature of animal shelters' work, involving several types of outcomes aside from adoption (such as transfer and euthanasia) is not properly addressed by current models [8].

The research can be considered important due to the following reasons. First, we compile and preprocess a dataset containing 148,317 records, unify the outcomes in accordance with a standardized taxonomy and build 34 features. Second, we create an ensemble model consisting of XGBoost, LightGBM and CatBoost as base models and logistic regression as meta-model. Third, we implement a SHAP analysis to examine the significance of the selected variables from the perspective of the model as a whole and in particular. Fourth, we perform the ablation study to measure the contribution of each component of the model architecture.

As the structure of the paper shows, chapter 2 deals with the literature review while chapter 3 describes our dataset. Next, chapter 4 introduces our proposed EEML methodology, and chapter 5 demonstrates experimental results. Chapter 6 discusses the obtained results and their limitations.

2. Related Work

2.1 Machine Learning for Animal Welfare Prediction

The use of machine learning for shelter outcome prediction started off with pilot research based on limited and single-institution datasets. Specifically, logistic regression was used by Reese et al. [3] to predict adoption in 30 days using a dataset of 4,200 cases from a shelter in Texas. In the process, the authors were able to identify age

and breed as major predictors while ignoring dynamic factors such as seasonality of intake and length of stay. Single-variable tests were also carried out by Patronek et al. [9], which led to the discovery that intake type, stray versus surrender, was a more accurate predictor compared to what was known up until then. Tree-based ensemble methods were next explored by Brown and Morgan [4], which involved applying random forest on a dataset of 12,500 cases from three California shelters and achieving an AUC of 0.79 in binary prediction of adoption. While this represented an improvement over linear models, the authors noted a substantial variance between the shelters, which they attributed to differences in intake and breed naming conventions. More recently, Yun et al. [10] came up with a gradient boosting model for the same problem based on Korean shelters' dataset. This time around, the model achieved an F1-score of 0.81 for the adopted class using 18 features and one-shelter training. Artificial neural networks have also been attempted in this application. Specifically, Salamon and Becker [11] found that the inclusion of photo-based visual features via multilayer perceptron increased the AUC score by 3.2 points. However, the costs and ethical considerations associated with image collection mean this approach can hardly be adopted by resource-poor shelters. Finally, transformer-based models relying on textual features extracted from intake notes were tried by Zhao et al. [12]; but, as expected, availability of such textual features is limited due to quality issues.

2.2 Ensemble Learning and Stacking Architectures

The theory of ensemble learning guarantees that blending of heterogeneous base models results in reduced error bias and variance compared to any constituent algorithm as long as their mistakes are not correlated [13]. In turn, boosting-based techniques such as XGBoost [14], LightGBM [15], and CatBoost [16] have been established as state-of-the-art methods for tabular data competition and benchmark tasks due to their superior performance compared to the best deep learning solutions on small to medium-sized tables. Stacked generalisation or stacking, which involves training a meta-model on the output of base estimators in out-of-fold setting, constitutes an extension of ensemble learning where the optimal weights for ensembling are found via data-driven learning rather than being fixed.

According to the recent literature reviews of ensemble methods in healthcare and social analysis [18, 19], stacking ensembles of gradient-boosted trees represent the most accurate solution for multiclass classification problems that involve mixed-type tabular data. In spite of that, stacking ensembles were never used in the context of animal shelters and no work on ensemble learning in that field employed cross-validation-based hyperparameter optimisation.

2.3 Explainable AI in Decision-Critical Domains

Explainable Artificial Intelligence (XAI) has emerged as an independent area of study primarily due to the shortcomings that emerge in the case of mere accuracy when AI models are put into action in practical environments [6]. One such method used in the field of explainability is SHAP (SHapley Additive exPlanations).

Introduced by Lundberg & Lee [20], SHAP uses principles from cooperative game theory to allocate values to input features according to their relative contributions towards predictions while satisfying some key properties of fairness. Subsequent applications of the method include clinical decision making [21], credit scoring [22] and even criminal risk assessments [23] among others and in all cases, it was found that the explanations generated passed the scrutiny of domain experts who found the explanations consistent with their pre-existing knowledge of the systems. In terms of animal welfare informatics, SHAP is not applicable since researchers used LIME (Local Interpretable Modelagnostic Explanations) on a binary classifier for shelter outcomes [24].

SHAP's TreeExplainer algorithm, by contrast, provides exact Shapley values for tree-based models in polynomial time [25], making it particularly well-suited to the gradient boosting architectures employed in the present study. Our work is the first, to our knowledge, to apply SHAP systematically to multiclass shelter outcome prediction, filling an important gap at the intersection of XAI and animal welfare informatics.

2.4 Summary of Research Gaps

Table 1 synthesizes the principal limitations identified in prior work and maps them to the contributions of the present study.

Table 1. Positioning of the proposed EEML framework relative to representative prior work.

Study	Multi-shelter	Multiclass	XAI	Ensemble	AUC/F1 Reported
Reese et al. [3]	×	×	×	×	Acc 67.4%
Brown & Morgan [4]	✓	×	×	RF only	AUC 0.79
Yun et al. [10]	×	Partial	×	GBM only	F1 0.81
Salamon & Becker [11]	×	×	×	×	AUC 0.83
Ramos-Soto et al. [24]	×	×	LIME	×	F1 0.78
Proposed EEML	✓	✓	SHAP	Stacking	AUC 0.931 / F1 0.847

1. Dataset and Preprocessing

The dataset was assembled from publicly archived shelter management exports and de-identified operational records provided under data-sharing agreements with three municipal shelters: Shelter A (North America, 2015–2022), Shelter B (Western Europe, 2016–2022), and Shelter C (East Asia, 2018–2022). After deduplication,

¹ 1.1 Data Sources and Ethical Considerations

removal of records with missing outcome labels, and harmonisation of species taxonomy to domestic cats and dogs—the two species for which cross-site sample sizes were sufficient—the combined corpus comprised 148,317 individual intake records. All animal identification information was removed prior to analysis; the study therefore poses no human-subjects ethical risk and was exempt from institutional ethics review under applicable national guidelines.

3.2 Outcome Taxonomy

The target variable was mapped to a harmonised four-class taxonomy following the taxonomy proposed by Karsten et al. [8]: (1) Adopted—animal placed in a private home via standard adoption; (2) Transferred— animal relocated to a partner organisation or rescue group; (3) Returned-to-Owner (RTO)—stray animal reclaimed by its original owner; and (4) Euthanised—animal humanely euthanised due to medical, behavioural, or capacity reasons. Table 2 presents the class distribution across the combined dataset.

Table 2. Outcome class distribution in the curated EEML dataset.

Outcome Class	Count	Percentage (%)	Shelter Sites
Adopted	71,192	48.0	A, B, C
Transferred	34,112	23.0	A, B, C
Returned-to-Owner	25,814	17.4	A, B
Euthanised	17,199	11.6	A, B, C
Total	148,317	100.0	—

3.3 Feature Engineering

Thirty-four features were engineered across five categories. Animal biometric features (7) include species, breed (grouped into 12 standardised categories per American Kennel Club and Fédération Cynologique Internationale taxonomies), age-at-intake, sex, reproductive status, coat colour, and body weight. Health and condition features (9) capture vaccination completeness, microchip status, presence of reported injury or illness, body condition score (BCS 1–9 scale), zoonotic disease flag, surgical history, and dental health score. Behavioural features (6) encode standardised temperament assessment scores for sociability with humans, conspecific compatibility, reactivity, trainability, energy level, and history of aggression. Shelter operational features (8) include intake type (stray, owner surrender, confiscation, transfer-in), intake season, weekday of intake, kennel location zone, concurrent shelter occupancy rate at intake, days since last distemper outbreak, staff-to-animal ratio, and shelter identifier. Derived temporal features (4) comprise cumulative length of stay, length-of-stay squared (to capture non-linear

decay in adoption probability [26]), interaction term between age and length-ofstay, and a rolling 30-day mean adoption rate at the time of intake.

3.4 Preprocessing Pipeline

The preprocessing pipeline comprised six sequential stages. Missing value imputation was performed using multivariate imputation by chained equations (MICE) for continuous features and mode imputation for nominal categorical features, with missingness rates ranging from 0.3% (microchip status) to 7.1% (body condition score). Breed standardisation resolved 342 distinct free-text breed entries to 12 canonical categories using a string-similarity matching algorithm with manual verification. Categorical encoding employed target encoding for high-cardinality nominal features (>10 levels) and one-hot encoding for low-cardinality features, following the recommendations of Micci-Barreca [27]. Continuous features were standardised using robust scaling (median and interquartile range) to mitigate the influence of outliers in weight and age distributions. Class imbalance—particularly affecting the euthanised class (11.6%)—was addressed via Synthetic Minority Oversampling Technique (SMOTE) applied exclusively within training folds to prevent data leakage [28]. Finally, multicollinearity was assessed using the variance inflation factor (VIF); three features with $VIF > 10$ were removed prior to model training.

4. Proposed EEML Framework

4.1 Framework Architecture

The proposed EEML structure has the two-level stacking architecture. At the first level, three different gradient-boosting algorithms are used as base models: XGBoost [14], LightGBM [15], and CatBoost [16]. All three algorithms have been chosen because of their unique inductive bias; XGBoost utilizes level-by-level decision trees with L1 and L2 regularization, LightGBM employs leaf-wise growing approach with efficient gradient-based one-side sampling, and CatBoost applies order-sensitive target encoding with symmetrical tree growing technique for mitigating categorical noises. Page 6 The empirical evaluation of correlation between predictions showed that Pearson correlations between predictions of all three models varied between 0.71 and 0.84.

Each base learner generates soft probability vectors (four-dimensional, one per outcome class) via out-of-fold prediction within a stratified five-fold cross-validation loop. Out-of-fold prediction ensures that the metalearner training data is never generated by a model trained on the same observations, preventing information leakage that would otherwise inflate meta-learner performance estimates [17]. The stacked meta-feature matrix— of dimension $N \times 12$, where N is the training set size and 12 represents four probability outputs from each of three base learners—is then used to train a multinomial logistic regression meta-learner with L2 regularisation. Logistic

regression was selected for the meta-learner to preserve interpretability at the combination stage and to avoid overfitting the relatively low-dimensional meta-feature space [17].

4.2 Hyperparameter Optimisation

Hyperparameter optimisation was conducted using Bayesian optimisation with a Tree-structured Parzen Estimator (TPE) as implemented in the Optuna framework [29], with 150 trials per base learner. The optimisation objective was macro-averaged F1-score evaluated on the held-out fold within the outer cross-validation loop. Key hyperparameters searched for each base learner included: learning rate (log-uniform, 1×10^{-3} to 3×10^{-1}), maximum tree depth (integer, 3–12), number of estimators (integer, 100–2000 with early stopping patience of 50 rounds), L1 and L2 regularisation strengths (log-uniform, 1×10^{-5} to 10), column subsampling ratio (uniform, 0.5–1.0), and row subsampling ratio (uniform, 0.5–1.0). For the final hyperparameter configuration was derived by computing the average of the best-performing values from all five outer folds, promoting robust and consistent model generalization.

4.3 SHAP Explainability Layer

Interpretability of models was measured by calculating the SHAP TreeExplainer values [25] of each base classifier independently, as well as for the ensemble predictions on probabilities of outcomes. Global interpretation of the model was performed by aggregating mean absolute SHAP values of all data samples and classes of outcomes in order to construct a general ranking of features. Beeswarm plots of class conditional SHAP values were constructed for each of the classes of outcomes, which allows us to understand the directionality of feature influence – that is, whether a higher value of the feature leads to an increased or decreased probability of the outcome. Consistency of explanations was tested by applying the explanation consistency measure proposed by Bhatt et al. [30]. Besides post-hoc explainability of predictions, we have also created a prototype explanation report that could be used by practitioners at animal shelters: a single-page PDF containing both predicted outcome probabilities and top five features affecting the outcome written in natural language (for example, 'Long length of stay decreases adoption probability by approximately 18%'). The document readability was measured with the Flesch Reading Ease measure, with a target of at least 60 points.

4.4 Evaluation Protocol

Model Evaluation of the models made use of a nested cross-validation strategy with an outer five-fold stratified partitioning for fair estimation of the performance and an inner three-fold partitioning for hyperparameter tuning. Class balance was preserved by means of stratification during the cross-validation process. The metrics that are presented hereafter refer to macroaveraged performance metrics since all four classes were weighted equally despite their different frequencies: F1-score, precision, recall, and AUC-ROC (one-vs-rest). Furthermore, Matthews Correlation Coefficient (MCC) was used as an additional metric that can be considered robust to class

imbalance [31]. Performance differences between models were compared by means of paired Wilcoxon signed rank test corrected via HolmBonferroni method for $\alpha = 0.05$.

5. Experimental Results

5.1 Overall Classification Performance

Table 3 presents nested cross-validation performance for all evaluated models. The proposed stacking ensemble (EEML) achieved the highest performance across all primary metrics: macro F1-score of 0.847 (± 0.009), AUC-ROC of 0.931 (± 0.005), precision of 0.851, recall of 0.843, and MCC of 0.812. All pairwise differences between EEML and individual base learners were statistically significant after Holm-Bonferroni correction ($p < 0.01$). The improvement of EEML over the best single base learner (LightGBM, $F1 = 0.796$) represents a margin of 5.1 percentage points, consistent with theoretical predictions of ensemble gain under moderate inter-learner diversity [13].

Table 3. Nested cross-validation performance of all evaluated models (mean $\pm$ std over 5 outer folds).

Model	Macro F1 Score	Macro F1 Score	Macro F1 Score	Macro F1 Score	Macro F1 Score
Logistic Regression	0.634 ± 0.014	0.634 ± 0.014	0.634 ± 0.014	0.634 ± 0.014	0.634 ± 0.014
Decision Tree	0.663 ± 0.019	0.663 ± 0.019	0.663 ± 0.019	0.663 ± 0.019	0.663 ± 0.019
Random Forest	0.754 ± 0.011	0.856 ± 0.009	0.761 ± 0.012	0.748 ± 0.011	0.712 ± 0.013
XGBoost	0.783 ± 0.010	0.881 ± 0.007	0.789 ± 0.011	0.778 ± 0.010	0.741 ± 0.012
LightGBM	0.796 ± 0.009	0.893 ± 0.006	0.801 ± 0.010	0.792 ± 0.009	0.756 ± 0.011
CatBoost	0.791 ± 0.010	0.889 ± 0.007	0.797 ± 0.011	0.787 ± 0.010	0.749 ± 0.012
Random Forest	0.754 ± 0.011	0.856 ± 0.009	0.761 ± 0.012	0.748 ± 0.011	0.712 ± 0.013
EEML (Proposed)	0.847 ± 0.009	0.931 ± 0.005	0.851 ± 0.009	0.843 ± 0.009	0.812 ± 0.010

5.2 Per-Class Analysis

Class-conditional performance analysis (Table 4) reveals that the EEML framework achieves its highest F1-score for the Returned-to-Owner class ($F1 = 0.891$), which benefits from highly discriminative features including intake type (stray) and microchip status. The Euthanised class, despite being the minority class (11.6%), achieves an F1-score of 0.803—a notable improvement over the best single model (LightGBM, $F1 = 0.741$), attributable to

SMOTE-balanced training and the ensemble's reduced variance. The Adopted class achieves $F1 = 0.862$, while the Transferred class records the lowest F1 at 0.831, reflecting the greater contextual dependency of transfer decisions on inter-organisational relationships not fully captured by the available features.

Table 4. Per-class F1-scores for the EEML framework versus best single-model baseline (LightGBM).

Outcome Class	EEML F1-Score	LightGBM F1-Score	Improvement (Δ)
Adopted	0.862	0.809	+0.053
Transferred	0.831	0.774	+0.057
Returned-to-Owner	0.891	0.843	+0.048
Euthanised	0.803	0.741	+0.062

5.3 SHAP Feature Importance Analysis

Global For all outcomes, the global SHAP analysis shows length of stay as the most critical feature (mean $|\text{SHAP}| = 0.312$) followed by intake type (0.287), age at intake (0.241), health condition score (0.198), and vaccination completeness (0.176). The first five features together explain 67.3% of model importance, suggesting moderate concentration of importance distribution. The age $\times$ length-of-stay interaction factor is sixth most important feature (mean $|\text{SHAP}| = 0.149$), thereby confirming the research hypothesis on non-linearity and acceleration of adoption probability decay rate with respect to increase in age due to staying long periods within shelters – a result which matches with the findings reported by Weiss et al. [26]. Further, class conditional SHAP analysis highlights crucial directional information on feature contributions. For Adopted class, low length of stay (positive direction), youthfulness, vaccination completeness, and high sociability score are the key positively contributing features. In contrast, for Euthanised class, aggressive intake behaviour (mean $|\text{SHAP}| = 0.231$), health condition flag (0.214), and high shelter occupancy at intake (0.187) are the key positively contributing factors, whereas owner-surrendered intake type and vaccination completeness status contribute negatively towards prediction. The results match well with expert domain knowledge as confirmed from structured interviews conducted with shelter directors from all three sites participating in the study.

5.4 Ablation Study

An ablation study was performed to measure the marginal effect of each building block on the final performance. Without class balancing using SMOTE, Euthanised-class F1 decreased by 8.6% to 0.734. This finding validates the need for a class-balancing technique in order to improve performance in relation to the minority class. Replacing Bayesian optimization with random search (same computation cost) decreased macro F1 by 2.1% (0.847 $\rightarrow$ 0.826). Using probability averaging in place of logistic regression (meta-learner) decreased macro F1 by 1.8% (0.847 $\rightarrow$ 0.829), which highlights the importance of learning combination weights through a meta-model.

Excluding CatBoost (keeping XGBoost & LightGBM) from the candidate list of base models led to a 1.4 point decrease in macro F1.

6. Discussion

6.1 Practical Implications for Shelter Management

The EEML framework predicts outcomes in ways that provide multiple practical uses for how shelters function. In the first instance, the model identifies the length of time an animal stays as the most influential factor. As the probability that an animal will be adopted decreases significantly after 14 days, shelters can choose to focus their promotional efforts on animals that have been in the facility for nearly two weeks. To address a second point, when an animal has all its vaccines, it is more likely to experience a positive result in every category. Because the model measures how much more likely an adoption becomes in mathematical terms, staff can justify spending money on medical procedures when animals first arrive even if they have little money available. By looking at a third factor, the model shows that the number of animals present in a shelter when a new animal arrives is very important for predictions. If managers monitor how many beds are full in real time and use this model, they are able to make choices about moving animals to other locations before a facility becomes too crowded. Due to those tools, people can share the animal population among different shelters in a more balanced way.

In a structured study of usability, researchers evaluated reports that explain data using SHAP. To perform this 24 people who work at shelters participated, where 12 individuals had no training in data literacy and 12 individuals had basic training. By using the System Usability Scale (SUS), the researchers measured how well the participants could use the system. Average SUS scores are 76.4 for the group without training and 84.1 for the group with training - those numbers are at a level that represents a range from good to excellent. On that account the results suggest that the interface for explanations is easy for many different types of users to understand and use. When the participants provided feedback in their own words, they emphasized that the SHAP summaries use simple language. And those summaries are more useful for making decisions than if the system only provided numbers for raw probability. If the results are considered together, they show that it is useful to include a layer for Explainable AI (XAI) within the system.

6.2 Limitations

Several there are multiple limitations in the current study that the authors acknowledge. In the first instance, the dataset is from three shelters in different geographic areas but it does not represent the many different ways that shelters operate globally - this is especially true for settings in the Global South that have few resources and where staff may not record data in digital formats. On the second point, the scores for how animals behave are vulnerable to differences in how various people observe and record them - those differences are not measured in the original records. If researchers perform future work, they should include protocols for behavior tests that are standard, like

SAFER or BARK and they should include coefficients for reliability. For the third point, the model is static and it does not account for how factors change over time. It is possible that a recurrent or temporal convolutional architecture is more effective, because it models the probability of adoption as a series of events over time. To address the fourth point, SHAP values are useful because they explain how tree-based models function on local and global levels but they are not used to show that one thing causes another. As an example the length of a stay is a predictor for a low probability of adoption but it is also a result of that low probability. And because this is a feedback loop, the model is not able to separate those factors without strategies for causal identification. Because of this future work is more likely to provide insights for policy if it uses frameworks for causal inference, like instrumental variable estimation or difference-in-differences designs.

7. Conclusion

The Ensemble Explainable Machine Learning (EEML) paradigm for predicting multiclass shelter outcomes was introduced in this research. The proposed system significantly outperforms previous single-model approaches by achieving a macro F1-score of 0.847 and AUC-ROC of 0.931 on a harmonized dataset of 148,317 records spanning three international shelters by integrating XGBoost, LightGBM, and CatBoost base learners within a stacking architecture, optimized via Bayesian hyperparameter search, and paired with SHAP-based explainability. The three most significant predictors worldwide, according to the SHAP study, are duration of stay, intake type, and age-at-intake. Class-conditional significance patterns that support domain expert knowledge and offer specific operational solutions are also identified. The three most significant predictors worldwide are duration of stay, intake type, and age-at-intake, with class-conditional significance patterns that support specific operational interventions and are consistent with domain expert knowledge.

The EEML system, which offers decision-support capabilities that are simultaneously accurate, comprehensible, and realistically deployable, is a significant step toward evidence-based shelter management. (i) Extension to other species and outcome classes; (ii) incorporation of longitudinal time-series modeling to capture within-stay trajectory dynamics; (iii) causal inference analyses to support policy evaluation; and (iv) randomized controlled evaluation of the impact of framework deployment on adoption and euthanasia rates at the shelter level are some potential future research directions. The preprocessing and training codes will be made publicly available upon

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